

Chapman and Chapman (1967a, 1967b, 1969) presented an even more compelling demonstration of the phenomenon of illusory correlation. The authors were puzzled by a controversy within the field of clinical psychology over the use of the Draw-a-Person Test. This is a psychodiagnostic test in which the client is asked to “draw a person” and the drawings are scored according to a number of dimensions (for example, whether the figure drawn is muscular, has atypical eyes, is childlike, is fat). Clinicians had reported strong correlations between some of the features of drawings and particular symptoms and behavioral characteristics (for example, atypical eyes as being drawn by suspicious clients; big heads as being drawn by intelligent clients). However, these reports were never confirmed by researchers studying the test itself.

In one study, Chapman and Chapman (1967a) gave undergraduates who were unfamiliar with the Draw-a-Person Test a series of 45 drawings that they randomly paired with symptoms allegedly displayed by the people who drew them. These undergraduates “discovered” the same correlations that clinicians had been reporting. Because the drawings and symptoms were randomly paired, it appeared the undergraduates shared with the clinicians a preexisting bias as to what relationships would be found in the data. That is, they “discovered” relationships they expected to find, even when no such relationships were there.

Variables that tend to be falsely associated typically seem to have some prior association in people’s minds (Chapman & Chapman, 1967b). On the surface, it seems to make sense that suspicious clients might draw wide-eyed figures: The wide eyes might be an artistic or symbolic representation of their suspiciousness. The point here is that the associations we bring to a situation often color our judgment to such an extent that we see them even if they are not there.

Hindsight Bias

Consider the following decision: You need to choose between declaring a psychology major and an economics major. You consult your own performance, goals, likes, and dislikes and have long discussions with faculty in both departments, majors in both departments, teachers from both areas who have instructed you, friends, parents, and relevant others. You finally decide to become an economics major, primarily because of your interest in the topics in your classes and also because you like the economics faculty so much.

A few months later you start to discover you like your economics classes less and less, and you find your psychology courses more interesting than you previously had. You reopen the decision about your major, spend another couple of weeks rethinking your goals and interests, and decide to switch majors to psychology. When you announce this decision to your best friend, she says, “Well, I knew this was going to happen. It was pretty much inevitable. You don’t seem the type to fit in with the other majors in that department, and also, given the stuff you said about last term’s assignments, I knew you wouldn’t like it for

long.” Other friends of yours also express little surprise at your latest decision, confiding that they “knew all along” that you would change your major.

How is it that you yourself didn’t foresee this inevitable change in majors? How is it that your friends could see into your future and you could not? In fact, one likely answer is that your friends are in error and that they are suffering from something called **hindsight bias**. Fischhoff (1982b) described this bias as a tendency to “consistently exaggerate what could have been anticipated in foresight” when looking back (in hindsight) on an event (p. 341). The idea is that once you know how a decision has turned out, you look back on the events leading up to the outcome as being more inevitable than they really were.

Fischhoff (1975) demonstrated hindsight bias experimentally in the following way. Participants were asked to read a passage similar to the one shown in Box 13-4. Passages described either historical events (as does the one in Box 13-4) or clinical case descriptions of people. All participants were asked to rate the likelihood of a number of possible outcomes of the event. Some

Box 13-4 Example of Hindsight Bias

Read the following passage, then answer the question below:

[1] For some years after the arrival of Hastings as governor-general of India, the consolidation of British power involved serious war. [2] The first of these wars took place on the northern frontier of Bengal where the British were faced by the plundering raids of the Gurkas of Nepal. [3] Attempts had been made to stop the raids by an exchange of lands, but the Gurkas would not give up their claims to country under British control, [4] and Hastings decided to deal with them once and for all. [5] The campaign began in November 1814. It was not glorious. [6] The Gurkas were only some 12,000 strong; [7] but they were brave fighters, fighting in territory well suited to their raiding tactics. [8] The older British commanders were used to war in the plains where the enemy ran away from a resolute attack. [9] In the mountains of Nepal it was not easy even to find the enemy. [10] The troops and transport animals suffered from the extremes of heat and cold, [11] and the officers learned caution only after sharp reverses. [12] Major-General Sir D. Ochterlony was the one commander to escape from these minor defeats.

In light of the information appearing in the passage, what was the probability of occurrence of each of the four possible outcomes listed below? (The probabilities should sum to 100%.)

- _____ British victory
- _____ Gurka victory
- _____ Military stalemate with no peace settlement
- _____ Military stalemate with a peace settlement

SOURCE: Fischhoff (1975, p. 289).

were additionally told which one of the outcomes had actually occurred and were asked to estimate (in hindsight) the probabilities of all the possible outcomes.

Results showed that participants who were told what the outcome actually was gave higher estimates of the probability that it would have happened, given the description in the passage, than did those who were not told what outcome had occurred. Surprisingly, participants saw as inevitable the outcome that they were told had happened, regardless of which outcome it was. Participants who were told that the British won saw British victory “all along” in the passage; those who were told that the Gurkas won saw that outcome as inevitable. Those who were told that the clash ended in a military stalemate also said that they saw that outcome coming.

How does hindsight bias apply to your friends in the hypothetical situation just described? Recall that they told you they “knew all along” that your decision to major in economics would not turn out well. It is likely, however, that your friends are looking back in hindsight, knowing how your original decision turned out, and are therefore more able to think of reasons that your decision turned out this way. Their ability to predict, in foresight, how your decision would turn out is probably far weaker. In short, to quote an old maxim, “Hindsight is always 20/20.” Recent investigations, by the way, have demonstrated the occurrence of hindsight bias in real-life contexts, with participants’ (mis)recollections of such events as the economic effects of the introduction of the Euro (Hoelzl, Kirchler, & Rodler, 2002), the impeachment verdict regarding President Clinton (Bryant & Guilbault, 2002), and the outcome of the O. J. Simpson trial (Demakis, 2002).

Confirmation Bias

In my hometown, parents can place their public school children in one of seven options for first grade. Parents of kindergartners, therefore, spend a lot of time and energy trying to find the “best” option for their child. Some parents do this by talking to other parents. For example, parents interested in the Spanish-immersion option may seek out other parents who have children enrolled in the immersion program, and ask if they like it. After a prospective parent has talked to, say, five happy Spanish-immersion parents, their own sense grows that the immersion program is right for their child.

What’s wrong with that, you may ask? Who better to tell a parent what the experience is like than another parent? Well, the name for the way in which the prospective parent is gathering information is called **confirmation bias**, discussed in Chapter 12. This is the tendency to search only for information that will confirm one’s initial hunch or hypothesis, and to overlook or ignore other information.

Parents go wrong if they *only* seek information that would potentially confirm their hunch that a particular option is the best. If they only talk to parents of children in the program, they talk to parents most likely to be happy customers of the program. (If they weren't happy with the program, then presumably those parents would have placed their children in other programs.) The most rational decision, then, would be made by talking to a randomly selected set of parents, or to parents who have transferred out of a particular option, as well as to parents with children still in that option.

Overconfidence

Consider the questions in Box 13-5, and choose from the two possible answers. After answering, give a rating of your confidence. If you have absolutely no idea of what the answer is, choose the value .5, to indicate that you think the odds you are right are 50–50. (Any number lower than .5 would indicate you think you are more likely to be wrong than right, so you should have chosen the other answer.) A rating of 1.00 means you are 100% certain your answer is correct. Values between .5 and 1.00 indicate intermediate levels of confidence, with higher numbers reflecting higher confidence.

Box 13-5 Some Trivia Questions

Choose one answer for each question, and rate your confidence in your answer on a scale from .5 (just guessing) to 1.0 (completely certain).

Which magazine had the largest circulation in 1978?

- a. Time
- b. Reader's Digest

Which city had the larger population in 1953?

- a. St. Paul, MN
- b. New Orleans, LA

Who was the 21st president of the United States?

- a. Arthur
- b. Cleveland

Which Union ironclad ship fought the Confederate ironclad ship Merrimack?

- a. Monitor
- b. Andover

Who began the profession of nursing?

- a. Nightingale
- b. Barton

For the purposes of this discussion, it matters very little how accurate your answers are. (If you simply *must* have the correct answers, they are b, a, a, a, a.) What matters here is the relationship between your accuracy and your confidence rating. In several studies (reviewed by Lichtenstein, Fischhoff, & Phillips, 1982), participants were given a long list of questions similar to those in Box 13-5. After they answered all the questions and gave confidence ratings,